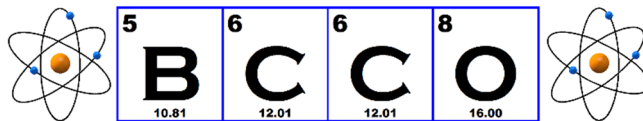


JUNIOR CANADIAN CHEMISTRY OLYMPIAD

British Columbia Chemistry Olympiad



2022

The Periodic Table/Data Sheet may be used for all parts of the contest, and non-programmable calculators are allowed.

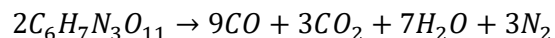
Write your answers on the Answer Sheet provided. Do not forget to write your name and your school name on the front of the Answer Sheet. While students are expected to attempt all questions for a complete examination in 1.5 hours, it is recognized that backgrounds will vary and students will not be eliminated from further competitions because they have missed parts of the paper.

- Which of the following chemical equations is properly balanced?
a) $C_6H_{12}O_6 + 12O_2 \rightarrow 6H_2O + 6CO_2$
b) $C_6H_8 + 6O_2 \rightarrow 5CO_2 + 4H_2O$
c) $C_{12}H_{22}O_{11} + H_2O \rightarrow 2C_6H_{12}O_6$
d) $C_2H_2 + 2O_2 \rightarrow 2CO_2 + H_2O$
e) $C_{12}H_{22}O_{12} + H_2O \rightarrow 2C_6H_{12}O_6$
- High Fructose Corn Syrup (HFCS) is used as a sweetener in soft drinks. The type of HFCS that is most commonly used in Coca-Cola is 55 percent fructose and 45 percent glucose (HFCS-55). The glucose and fructose concentrations in Coca-Cola are 0.270 M and 0.330 M respectively. What is the mass of glucose ($C_6H_{12}O_6$) and fructose ($C_6H_{12}O_6$) in a 375 mL can of Coke?
a) 20.5 g of fructose and 16.8 g of glucose
b) 22.3 g of fructose and 18.2 g of glucose
c) 59.5 g of fructose and 48.6 g of glucose
d) 33.0 g of fructose and 27.0 g of glucose
e) 27.5 g of fructose and 22.5 g of glucose
- The density of sodium chloride at 25 °C is 2.17 g/cm³. What is the volume of 1 mole of NaCl at 25 °C?
a) 26.9 cm³ b) 18.0 cm³ c) 126.8 cm³ d) 29.6 cm³ e) 58.4 cm³
- The New Afton mine in British Columbia is found to contain 0.68 grams of gold per ton of ore. If 45.7 tons of ore is collected from the New Afton mine, how many moles of gold can be extracted from the ore?
a) 0.015 b) 1.00 **c) 0.158** d) 6.12 e) 0.232
- Several general types of chemical reactions can occur based on what happens when going from reactants to products. What is the type of the following reaction?
$$CaCO_3 \rightarrow CaO + CO_2$$

a) Synthesis d) Neutralization
b) Single replacement e) Combustion
c) Decomposition
- The cause of anemia, a low level of red blood cells, is often iron deficiency. Pharmaceutical-grade ferric citrate can be prepared from the reaction of a ferric salt with citric acid. What is the iron concentration of a solution prepared with 1.30 g of iron chloride hexahydrate ($FeCl_3 \cdot 6H_2O$) in a 100 mL volumetric flask?
a) 4.81×10^{-1} M d) 8.01×10^{-2} M
b) 8.01×10^{-1} M **e) 4.81×10^{-2} M**
c) 4.81×10^{-3} M

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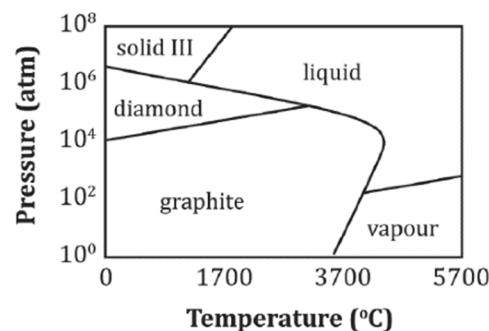
- 7) Nitrocellulose (also called guncotton) is a highly flammable compound that can be made by exposing cotton to a mixture of sulfuric and nitric acid. Guncotton can detonate if it is confined, as it would be in the barrel of a gun. The detonation reaction involving nitrocellulose is schematized below:



If 100 g of guncotton is detonating, how many grams of water is produced?

- a) 21.2 b) 6.06 c) 42.4 d) 350 e) 63.1
- 8) A 50.0 mL canister of Freon-12 (CF_2Cl_2) was heated in boiling water (100.0 °C) until the canister burst. If the canister was not defective, and had a burst rating of 102.3 atm, what minimum amount of Freon-12 was in the canister? (assume ideal gas behavior)
- a) 199 g b) 0.20 g c) 20.2 g d) 19.7 g e) 75.4 g
- 9) Which of the following molecules can form a hydrogen bond?
- a) NaH b) NH_3 c) CH_3F d) H_2CO e) CH_2Cl_2
- 10) What is the molecular geometry of PCl_5 ?
- a) Trigonal Planar
b) Trigonal Bipyramidal
c) Square Pyramidal
d) Trigonal pyramidal
e) Tetrahedral
- 11) Which of the following element is expected to be the most electronegative?
- a) Cl b) C c) Na d) S e) Br

- 12) Synthetic diamonds have many uses in industry. Diamond is the hardest known naturally occurring material. This property makes diamond the ideal material for machine tools and cutting tools. Synthetic diamonds can be obtained directly from graphite (carbon) under high pressure. The phase diagram of carbon is given below. What phase change occurs by going from 4700 °C to 2700 °C at a pressure of 10 atmospheres?



- a) Fusion
b) Deposition
c) Vaporization
d) Freezing
e) Sublimation

- 13) Effective nuclear charge (Z_{eff}) is a useful tool for understanding periodicity. Z_{eff} is the net positive charge experienced by an electron in a polyelectronic atom. The term "effective" is used because the positive charge of the nuclei cannot reach a given electron completely, due to the shielding effect of other electrons. Z_{eff} is a theoretically calculated value (not an atomic property) but a good estimate of Z_{eff} can be obtained using Slater's rule. Slater's rule states that $Z_{\text{eff}} = Z - S$ where Z is the atomic number of the atom and S a screening constant. In the second row of the periodic table, the screening value is 0.85 for a core electron and 0.35 for a valence electron. The screening constant (S) is obtained by the summation of all the screening values of other electrons (excluding the electron for which the effective nuclear charge is to be determined). Use Slater's rule to estimate the effective nuclear charge for a valence electron in nitrogen (N).
- a) 3.80 b) 5.60 c) 3.55 d) 3.90 e) 5.25

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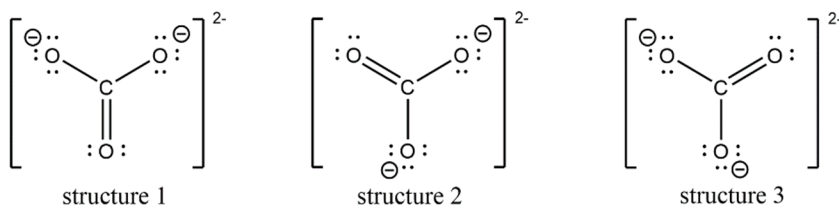
- 14) Calculate ΔH° for the reaction of ice with sodium metal at 0°C (assume the enthalpies given in the table are valid at this temperature):



$\Delta H_f^\circ [\text{Na}^+\text{(aq)}]$	-240.1 kJ/mol
$\Delta H_f^\circ [\text{OH}^-\text{(aq)}]$	-230.0 kJ/mol
$\Delta H_f^\circ [\text{H}_2\text{O(l)}]$	-285.83 kJ/mol
$\Delta H_{\text{fus}} [\text{H}_2\text{O(s)}]$	6.088 kJ/mol

- a) -185.1 kJ/mol
b) -178.2 kJ/mol
 c) 185.1 kJ/mol
 d) 178.2 kJ/mol
 e) -749.8 kJ/mol

- 15) Which Lewis structure(s) below is (are) good for carbonate?



- a) Structure 1 only
 b) Structure 2 only
 c) Structure 3 only
d) All the structures shown
 e) None of the structures shown

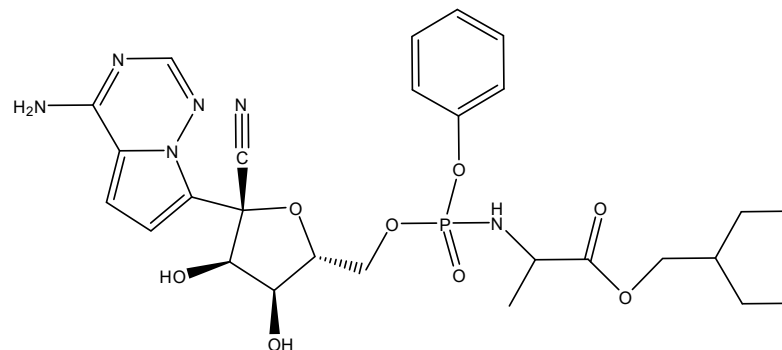
- 16) Which of the following atoms or ions has the smallest radius?

- a) Xe b) Fr^+ **c) Cs^+** d) Rn e) I^-

- 17) Which of the following atoms has the largest first ionization energy?

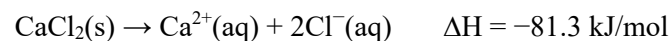
- a) Si b) Ca c) K d) Be **e) Cl**

- 18) The use of antiviral drugs to treat the coronavirus disease (COVID-19) has been controversial. Despite the controversy, the antiviral drug shown below was approved by the FDA for the treatment of COVID-19. The drug was also approved in Canada a few months later. Which of the following molecular formula is the correct one for the antiviral drug shown below?



- a) $\text{C}_{27}\text{H}_{34}\text{N}_5\text{O}_8\text{P}$ **d) $\text{C}_{27}\text{H}_{35}\text{N}_6\text{O}_8\text{P}$**
 b) $\text{C}_{25}\text{H}_{34}\text{N}_6\text{O}_8\text{P}$ e) $\text{C}_{27}\text{H}_{35}\text{N}_6\text{O}_7\text{P}$
 c) $\text{C}_{27}\text{H}_{32}\text{N}_6\text{O}_8\text{P}$

- 19) 25.0 g of calcium chloride (CaCl_2) is added to 50.0 g of ice at 0°C in an insulated container. Use the data below to calculate the final temperature of the system, in $^\circ\text{C}$.

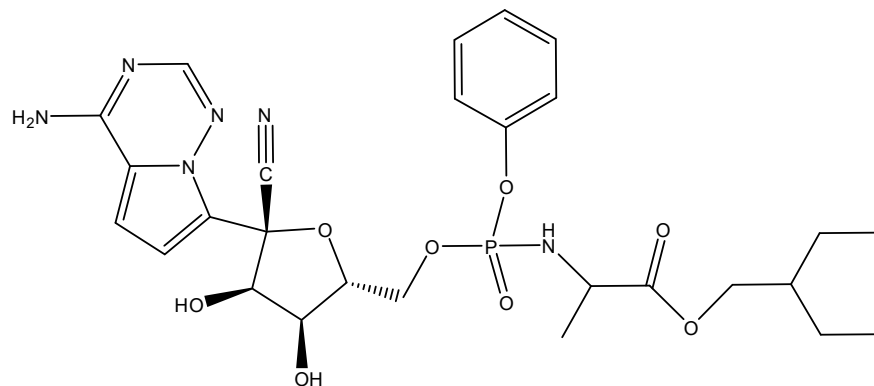


$C_{p,m} \text{H}_2\text{O(l)}$	$75.38 \text{ JK}^{-1}\text{mol}^{-1}$
$C_{p,m} \text{H}_2\text{O(s)}$	$37.66 \text{ JK}^{-1}\text{mol}^{-1}$
$\Delta H_{\text{fus}} [\text{H}_2\text{O(s)}]$	6.00 kJ/mol

- a) 15.7 b) 58.8 c) 21.9 **d) 7.9** e) 87.5

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20) The antiviral drug in question 18 has the structure shown as,



Which of the following correctly lists the organic functional groups presented in the molecule?

- Amine, Ester, Ether, Alkene, Nitrile
 - Amide, Ketone, Ether, Alkene, Nitrile
 - Ester, Alcohol, Amine, Nitro, Ether
 - Ketone, Cyanide, Amide, Benzene, Alcohol
 - Amine, Ether, Alcohol, Ester, Nitrile**
- 21) Using your knowledge of organic chemistry, structure of molecule and intermolecular forces to determine which of the following organic compounds has the highest vapor pressure under 1 atm and 20°C.
- Amino-butane
 - Butanol
 - Diethyl ether**
 - Ethanol
 - Ethyl Ethanoate
- 22) Canada is one of the world's largest producers of uranium used in the nuclear power plants. ^{238}U has a half life of 4.468×10^9 y. What would be the rate constant when ^{238}U undergoes radioactive decay?
- $1.55 \times 10^{-10} \text{ y}^{-1}$**
 - $1.99 \times 10^{19} \text{ y}^2$
 - $2.24 \times 10^9 \text{ y}$
 - $2.24 \times 10^{-10} \text{ y}^{-1}$
 - $6.68 \times 10^4 \text{ y}^{1/2}$
- 23) A solution is made by mixing 8.30ml of 0.630M NaCl, 4.20ml of 0.432M CaCl_2 , 5.10ml of 0.322M of FeCl_3 and 2.40ml of 0.280M of AgNO_3 . The final concentration of chloride ion in the mixture will be the closest to,
- 0.312M
 - 0.434M
 - 0.494M
 - 0.656M**
 - 0.689M
- 24) A company wants to electroplate a spoon with silver metal. What current is needed to electroplate 5.3 g of silver metal from a silver nitrate (AgNO_3) solution if the process runs for 45 minutes?
hint: use the equation $t = \frac{Q}{I}$ where Q is the amount of electricity in coulombs, I is the current in ampere and t is the time in seconds.
- 1.755 A**
 - 0.570 A
 - 1.115 A
 - 0.897 A
 - 105.3 A
- 25) A chemist adds 100 mL of water to 100 mL of a 0.35 M acetic acid solution ($\text{pK}_a = 4.80$). Calculate the pH of the resulting solution.
- 2.63
 - 2.78**
 - 5.56
 - 5.26
 - 2.98

End of examination questions.

	5 B 10.81	6 C 12.01	6 C 12.01	8 O 16.00	
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